

ML Hackathon - Participant Quickstart

Shared-Set Package Overview & Getting Started Guide

Welcome to the Network Intrusion Detection Machine Learning Hackathon!

This offline package contains all initial datasets, documentation, starter code, and submission tools necessary for the 5-hour hackathon.

Package Directory Structure

- **data/**: Contains train.csv (60%) and validation.csv (15%). Both files have 40 columns (38 features + attack_cat + Label).
- **docs/**: feature_dictionary.pdf (full 38-feature reference) and evaluation_rules.pdf (rules, rubric, model submission guide).
- **starter/**: baseline.py (executable ML pipeline with threshold tuning), evaluate.py (submission checker), and predict.py (inference template).
- **submission/**: submission_template.csv (format specification for submissions).

Quickstart Instructions

1. Review docs/feature_dictionary.pdf to understand the 38 traffic flow features.
2. Execute the starter baseline: `cd starter && python baseline.py`.
3. Validate predictions locally using evaluate.py.
4. Prepare your model submission package (model.joblib + predict.py) as described in docs/evaluation_rules.pdf.
5. Prepare to receive challenge.csv at Hour 4 and final_test.csv at Hour 5.